TMIS — Temporal Memory Integrity Standard

OOF™ Origin Open Foundation™

Independent Methodological Authority

OriginID: OOF-OID-MEM-TMIS-2026-06-08-0005
Architecture Ecosystem: Structured Reality Standards™
Architecture Family: Memory Governance Intelligence Architecture (MGIA™)
Operational Layer: Temporal Memory Governance Layer
Governed Space: Temporal Memory Integrity
Category: AI & Interpretation
Subcategory: Temporal Memory Governance Architecture
Type: Parent Standard
Version: 1.0
Status: Canonical · Open Standard
Origin Date: 8 June 2026
Compatibility: OOF Methodology OS · Memory Integrity Standard (MIS) · Agent Memory Integrity Standard (AMIS) · User Memory Integrity Standard (UMIS) · Group Memory Integrity Standard (GMIS) · Temporal Reasoning Integrity Standard (future) · INTEGROS® — Integrity Standard
AI-Readable: Yes
Authority: OOF
Protection: MIP — Methodological Intellectual Property
Canonical Language: English (UCL)

Canonical Definition System

Canonical Definition

Temporal Memory Integrity Standard (TMIS) defines the structural conditions under which memory remains correctly associated with time, sequence, duration, chronology, temporal context, temporal relationships, and historical continuity throughout memory creation, storage, retrieval, modification, preservation, and utilization processes.

TMIS governs temporal memory.

The standard establishes the integrity conditions required for preserving the temporal dimension of memory across human, artificial, organizational, collective, and future intelligence environments.

Memory does not exist outside time.

Every memory has a temporal position.

TMIS governs the integrity of that temporal position.

A. Standard Abstract

Memory without time loses meaning.

Events happen in sequence.

Experiences occur in time.

Decisions are made at specific moments.

Knowledge evolves through time.

Context changes through time.

If temporal relationships are lost:

- causality becomes unclear
- chronology becomes distorted
- historical reconstruction becomes unreliable
- memory continuity degrades
- reasoning quality deteriorates

TMIS exists to govern these conditions.

B. Core Principle

Temporal memory becomes valid only when memory remains correctly associated with chronology, sequence, duration, temporal context, historical continuity, and operational reality throughout the memory lifecycle.

C. Scope

This standard may apply to:

- AI memory systems
- autonomous agents
- Human-AI environments
- organizational memory systems
- collective memory systems
- historical archives
- operational knowledge systems
- future intelligence ecosystems

TMIS applies wherever memory contains temporal meaning.

D. Why This Standard Exists

Memory is not merely information.

Memory contains:

- temporal order
- historical position
- sequence relationships

- continuity relationships
- causality relationships
- evolution relationships

Without temporal integrity:

- events may appear in the wrong order
- causes may appear after effects
- historical continuity may collapse
- decision histories may become unreliable
- memory reconstruction may fail

TMIS exists because time itself is a governable memory space.

E. Temporal Memory Integrity Logic

Temporal memory integrity exists only when the following remain materially preservable:

1. Chronological Integrity
Memory remains associated with correct sequence.

2. Temporal Context Integrity
Memory remains connected to its original temporal environment.

3. Historical Continuity Integrity
Memory remains linked across time.

4. Temporal Reconstruction Integrity
Historical timelines remain reconstructable.

5. Temporal Accountability Integrity
Temporal modifications remain accountable and reviewable.

F. Operational Architecture Space

TMIS defines the operational architecture space for:

- temporal memory governance
- chronology preservation
- historical continuity
- timeline reconstruction
- temporal context governance
- sequence integrity
- causality preservation
- temporal accountability

This space exists because memory derives meaning from temporal position.

TMIS governs whether that temporal meaning remains valid.

G. Difference Between Memory and Temporal Memory

Memory governs what is remembered.

Temporal memory governs when it happened.

MIS governs memory integrity.

TMIS governs temporal integrity of memory.

Both are required for trustworthy historical continuity.

H. Runtime Position

TMIS operates across all memory layers.

It provides temporal integrity conditions supporting:

- AMIS
- UMIS
- GMIS
- MMIS
- CMIS
- MEIS
- MGS

TMIS acts as the temporal governance layer of MGIA™.

I. Temporal Memory Failure Rule

Temporal memory failure occurs when memory loses its temporal validity.

Examples include:

- timeline corruption
- sequence inversion
- temporal drift
- historical distortion
- chronology loss
- causality confusion

TMIS exists to expose and govern these conditions.

J. Validity Logic

A temporal memory environment is valid under TMIS when:

- chronology remains correct
- sequence remains reconstructable
- temporal context remains identifiable
- historical continuity remains preservable
- causality remains understandable
- temporal accountability remains visible

A temporal memory environment becomes invalid when:

- chronology collapses
- timelines become unreliable
- sequence becomes corrupted
- temporal relationships disappear
- historical reconstruction becomes impossible

K. Relationship to Other OOF Standards

TMIS derives from:

Memory Integrity Standard (MIS)

and supports:

- AMIS
- UMIS
- GMIS
- MEIS
- CLIA®
- MGIA™

TMIS governs the temporal dimension of memory.

Other memory standards govern the memory objects themselves.

L. Foundational Principle

Memory may survive without temporal integrity, but historical meaning, causality, continuity, and reconstruction progressively disappear.

Canonical Closing Statement

Temporal Memory Integrity Standard (TMIS)
defines the structural conditions under which memory
remains correctly associated with time, sequence, duration, chronology, temporal context,
temporal relationships, and historical continuity throughout memory creation, storage, retrieval,
modification, preservation, and utilization processes.

Memory derives meaning from time.

Temporal memory integrity therefore becomes
a foundational condition of trustworthy
historical continuity, chronology, causality,
and memory reconstruction across intelligence systems.

Module Architecture

→ CIM — Chronological Integrity Module

→ TCIM — Temporal Context Integrity Module

→ HCIM — Historical Continuity Integrity Module

→ TRIM — Temporal Reconstruction Integrity Module

→ TAIM — Temporal Accountability Integrity Module

Related Documents

→ About the Temporal Memory Integrity Standard (TMIS)

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Canonical Source

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